Seattle Infinity Math Circle TST **September 20th, 2026**

Computational Round **75 minutes**

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1. Three ants, Christopher, Kevin, and Raymond, crawl at the same speed on the Cartesian plane. Christopher starts at $(0, 0)$ and each second moves either one unit up or right; Kevin starts at $(2, 8)$ and moves down or right; Raymond starts at $(9, 1)$ and moves up or left. If all three eventually meet at the same point without any two meeting earlier, how many possible sets of paths are there?

2. Let p_1, p_2, $\ldots$, p_8 be distinct primes such that for all i,

$$p_i^2 - p_i + 1 \equiv 0 \pmod 7$$
$$p_1 + p_2 + \cdots + p_8 \equiv 2 \pmod 7.$$

   How many positive divisors of $N = p_1 p_2 \ldots p_8$ are congruent to 1 mod 7?

3. Let $\theta = 2\pi/2027$. Define

$$A = \sum_{n=0}^{675} \cos(3n\theta).$$

   Compute

$$\frac{1}{A} - \frac{1}{\cos\theta}.$$

4. Let $\triangle ABC$ be an acute triangle with orthocenter H and circumcenter O. Let the midpoint of $\overline{BC}$ be M. If $MO = 6$, $MH = 7$, and $MA = 19$, compute BC.

5. Kevin plays a game on the number line. Each second, if at position n, he gains n points, then moves to $n - 1$ or $n + 1$ with equal probability. The game ends when Kevin reaches 0 or 15.

   If he starts at position 6 with 0 points, what is Kevin's expected final score?

6. Christopher constructs a random sequence of real numbers $a_1, a_2, a_3, \ldots$ as follows. Let $a_1 = \frac{3}{4}$. For each $n \geq 2$, he independently chooses a_n to be $\frac{3}{4}|a_{n-1}|$ with probability $\frac{2}{3}$ and $-a_{n-1}^2$ with probability $\frac{1}{3}$. Find the expected value of $\sum_{n=1}^{\infty} a_n$.

7. Find the number of ordered pairs of positive integers (x, y) with $x, y \leq 506$ such that

$$x^y \equiv y^x \pmod{23}.$$

8. A cyclic hexagon $ABCDEF$ satisfies $AB = BC = 1$, $CD = DE = \sqrt{2}$, and $EF = FA = \sqrt{5}$. Let X be a point in the interior of the circumcircle. Find the minimum possible value of $XA + XB + XC + XD + XE + XF$.

9. Ian and Raymond are playing a guessing game. Ian chooses a number $a \in [0, 1]$. Initially, $s = 0$ and $t = 1$. On each turn, Raymond guesses uniformly at random from $[s, t]$. If his guess exceeds a, he sets t equal to his guess; otherwise, he sets s equal to his guess. This process repeats. If $a = \frac{1}{3}$, find the expected value of Raymond's 9th guess.

10. Points A, B, C, D, E lie on a circle in that order with $AB = AE = 7$, $BC = CD = 3$, and $AC = 8$. Let $\overline{BD}$ intersect $\overline{CE}$ at X. Compute CX/XE.

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Proof Round **150 minutes**

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1. **[5]** Let $\triangle ABC$ be a scalene triangle. Let D be the center of the circle passing through A tangent to $\overline{BC}$ at B. Let X be the projection of D onto the A-altitude of $\triangle ABC$, and let Y be the intersection of the angle bisector of $\angle ADB$ and the A-altitude of $\triangle ABC$. Prove that the midpoint of $\overline{XY}$ is also the midpoint of the A-altitude.

2. **[5]** Call a positive integer "d-amazing" for some digit $2 \leq d \leq 9$ if:

   - It can be expressed as a^b for some positive integers $a, b > 1$.
   - Every digit in the number is one of 0, 1, or d, and all 3 digits appear at least once.
   - The units digit is nonzero.

   Prove that there exist infinitely many d-amazing integers for each $d \in \{2, 3, 4, 5, 8\}$.

3. **[6]** Let a, b, and c be pairwise relatively prime positive integers. Prove that

   $$\gcd(a^3 + bc, b^3 + ac, c^3 + ab)$$

   divides $(a^2 - 1)(b^2 - 1)(c^2 - 1)$.

4. **[7]** Find all real numbers c for which there exists a nonzero function $f \colon \mathbb{R} \to \mathbb{R}$ such that

   $$f(x)^2 + cf(y)^2 = (f(x + y) + f(y))(f(x - y) + f(y))$$

   holds for all real numbers x and y.

5. **[8]** Lumpy and Champy play on the infinite square grid of lattice points, with edges joining adjacent points. Initially, only the origin is black. On Lumpy's turn, he deletes any edge. On Champy's turn, she colors black any white point adjacent to a black point by an undeleted edge. They alternate turns. The game ends when Champy has no legal move. Her score is the maximum taxicab distance between two black points, where the taxicab distance between (a, b) and (c, d) is $|a - c| + |b - d|$. Lumpy plays to minimize this score. Prove that:

   (a) If Champy moves first, Lumpy can limit Champy's score to at most 16.

   (b) If Lumpy moves first, Champy will always end with a score of at least 6.

6. **[10]** Let $P(x)$ be a cubic polynomial such that $P(x) = x$ and $P(P(x)) = x$ each have exactly 3 distinct real solutions. Prove that for all positive integers n, $P^n(x) = x$ has exactly 3 distinct real solutions.

7. **[11]** Distinct points A, B, C, A', B', and C', not all coplanar, are chosen on a sphere centered at O such that AA', BB', and CC' concur at a point $P \neq O$. Prove that line OP is perpendicular to the line formed by intersecting the planes of $\triangle ABC$ and $\triangle A'B'C'$.

8. **[12]** Mr. Least traps $N \geq 2$ contestants in his basement. The winner receives a prize pool, which is initially empty. Each day, while at least two contestants remain, Mr. Least assigns each contestant to either the *\$1 room* or the *\$2 room*, then selects one contestant uniformly at random. If the selected contestant is in the \$1 room, he adds \$1 to the prize pool and eliminates everyone in the \$2 room; if they are in the \$2 room, he adds \$2 and eliminates everyone in the \$1 room.

   Mr. Least wants to minimize the expected prize pool. Find an optimal strategy and determine for which N there is a unique optimal strategy.